CPPC 18/L

COMPUTER NETWORK & SECURITY

ACTIVITY No.3

Pinging

Submitted by:

LOGATOC, APRIL FOILA JOY M.

BS CpE 3B

Submitted to:

ENGR. JAN ERROL MAMPUSTI

Subject Professor

ACTIVITY 3: Cisco Pinging

Objectives

This activity aims to:

- Check if the devices connected in one local area network can communicate with each other.
- Understand how the ping command is used to test network connection.
- Observe how data packets are sent and received between computers in a network.
- Practice assigning IP addresses to devices in Cisco Packet Tracer.

Materials

The following materials were used for this activity:

- Three computers or PCs
- One network switch
- Ethernet cables
- Cisco Packet Tracer software

Procedure

Step 1: Network Setup

First, I placed three PCs and one switch in the Cisco Packet Tracer workspace. Each PC was connected to the switch using Ethernet cables. After connecting the devices, I checked if the cable connections were active and properly linked.

Step 2: IP Address Configuration

After setting up the devices, I assigned an IP address to each PC. All computers were configured under the same network so they could communicate with each other.

Device	IP Address	Subnet Mask
PC1	192.168.1.1	255.255.255.0

PC2	192.168.1.2	255.255.255.0
PC3	192.168.1.3	255.255.255.0

Since all PCs used the same subnet mask and network address, they were able to belong to one local area network.

Step 3: Testing the Connection

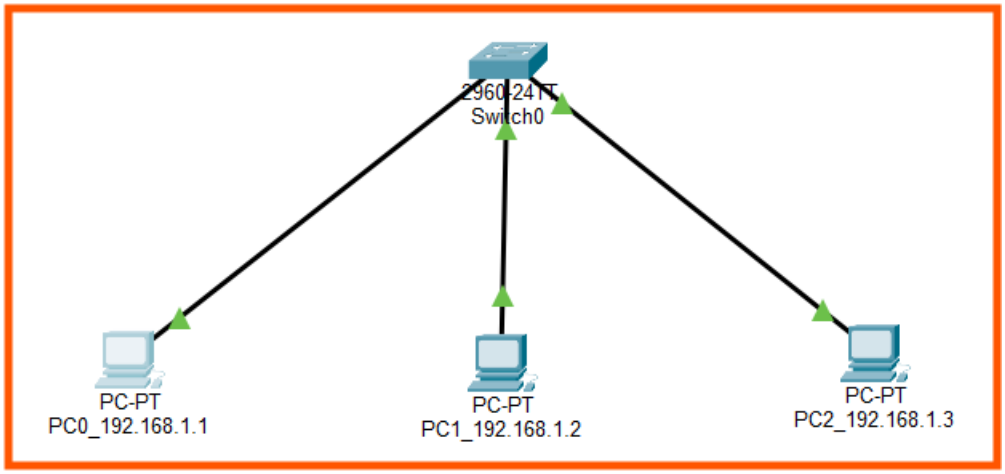
To test the connection, I opened the Command Prompt of each PC and used the ping command. For example, from one computer, I typed: ping 192.168.1.2

Then, I also tested the connection to the other PC by using its IP address. The same process was repeated on the other computers to make sure that all devices could communicate with one another.

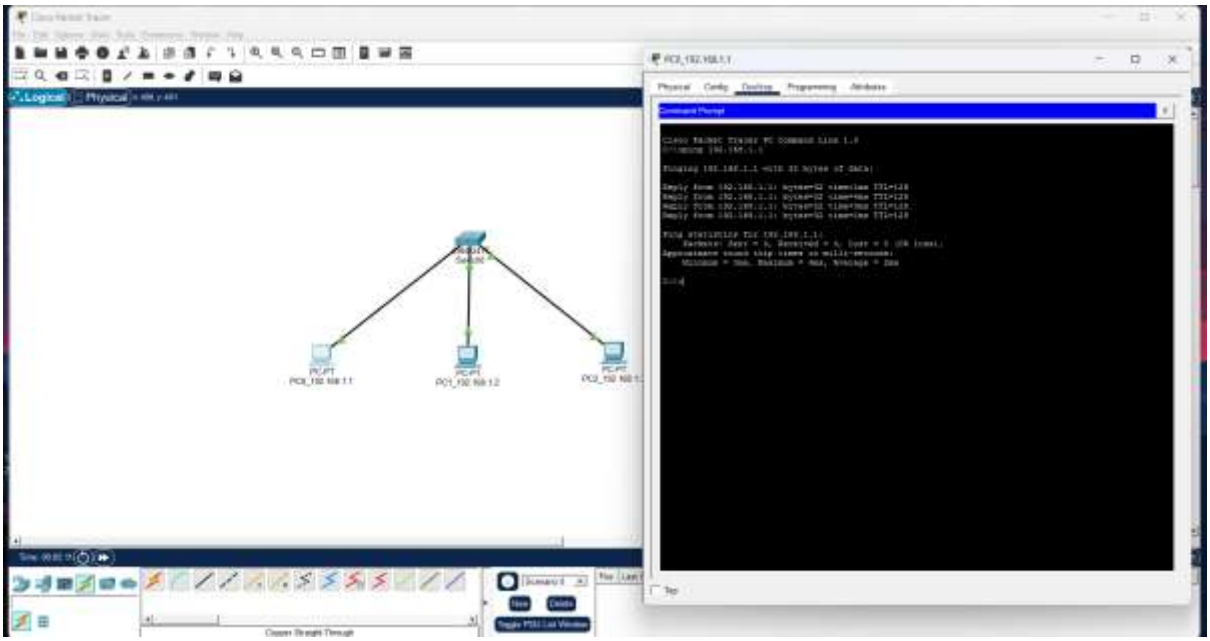
Conclusion

In this activity, I learned that devices connected to the same switch can communicate with each other as long as they are configured with the correct IP addresses and subnet mask. The ping command is useful because it helps verify whether a device can reach another device in the network. Overall, the activity showed the basic process of testing connectivity in a local area network using Cisco Packet Tracer.

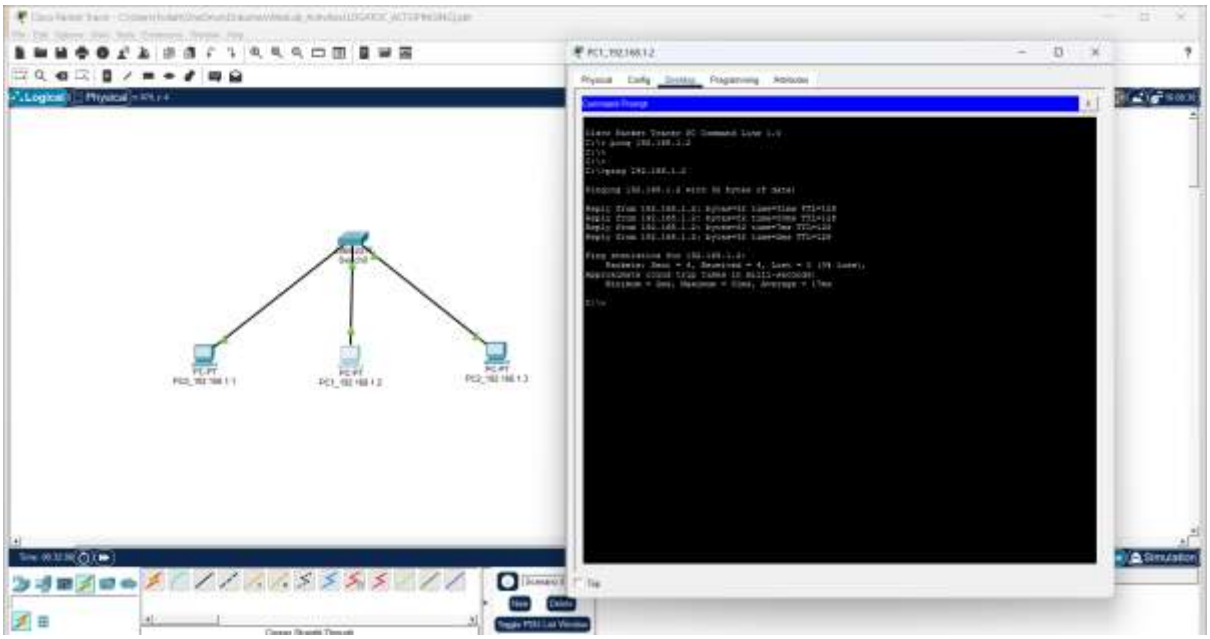
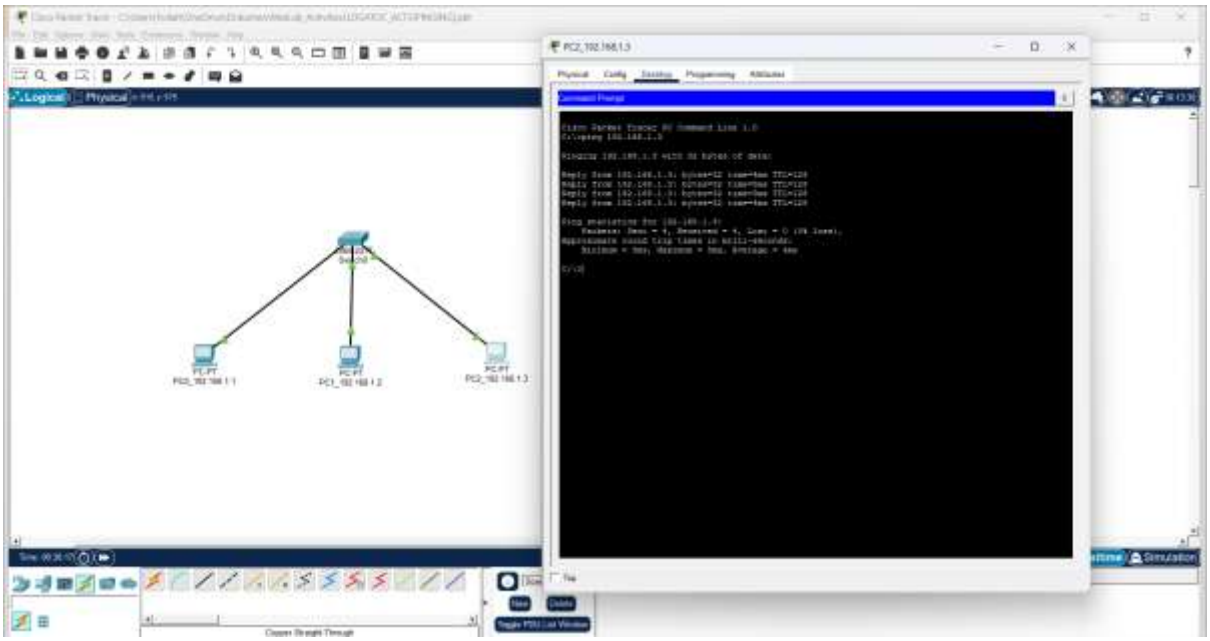
Documentation:



Ping at PC0_192.168.1.1



Ping at PC1_192.168.1.2



Ping at PC2_192.168.1.3